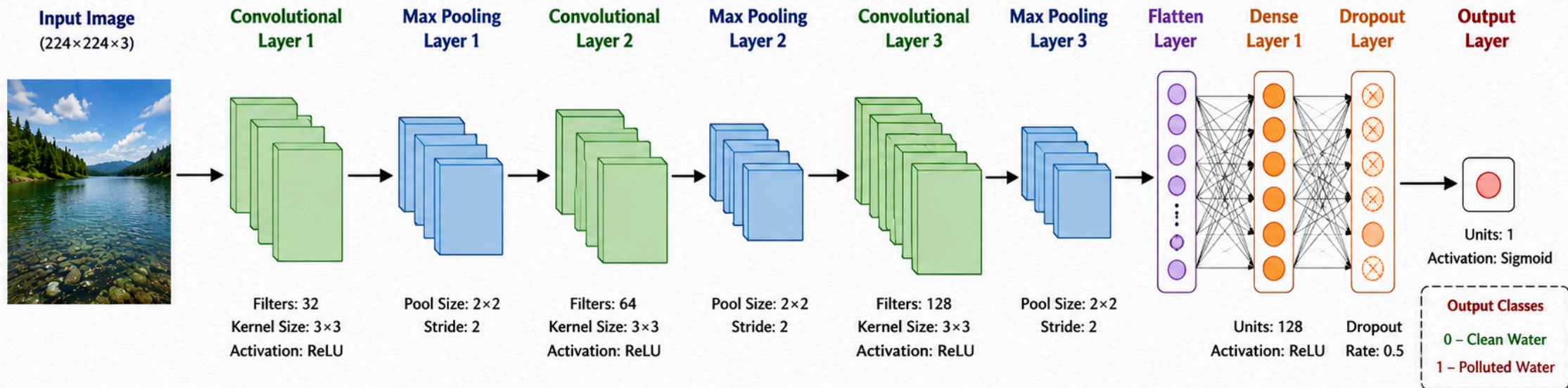


Proposed CNN Architecture for Water Pollution Classification



Input Layer	Conv2D + ReLU	Max Pooling	Conv2D + ReLU	Max Pooling	Conv2D + ReLU	Max Pooling	Flatten Layer	Dense Layer	Dropout Layer	Output Layer
224 × 224 × 3 RGB Image	32 Filters 3 × 3 Kernel	2 × 2 Pool Stride 2	64 Filters 3 × 3 Kernel	2 × 2 Pool Stride 2	128 Filters 3 × 3 Kernel	2 × 2 Pool Stride 2	1D Feature Vector	128 Units ReLU	Dropout Rate = 0.5	1 Unit Sigmoid

- ReLU activation is used in convolutional and dense layers for non-linearity.
- Max pooling reduces spatial dimensions and computational complexity.
- Dropout helps prevent overfitting and improves generalization.
- Sigmoid activation in output layer performs binary classification.